“Cloth Mate” Smart Automated Cloth Line

by Group P-18

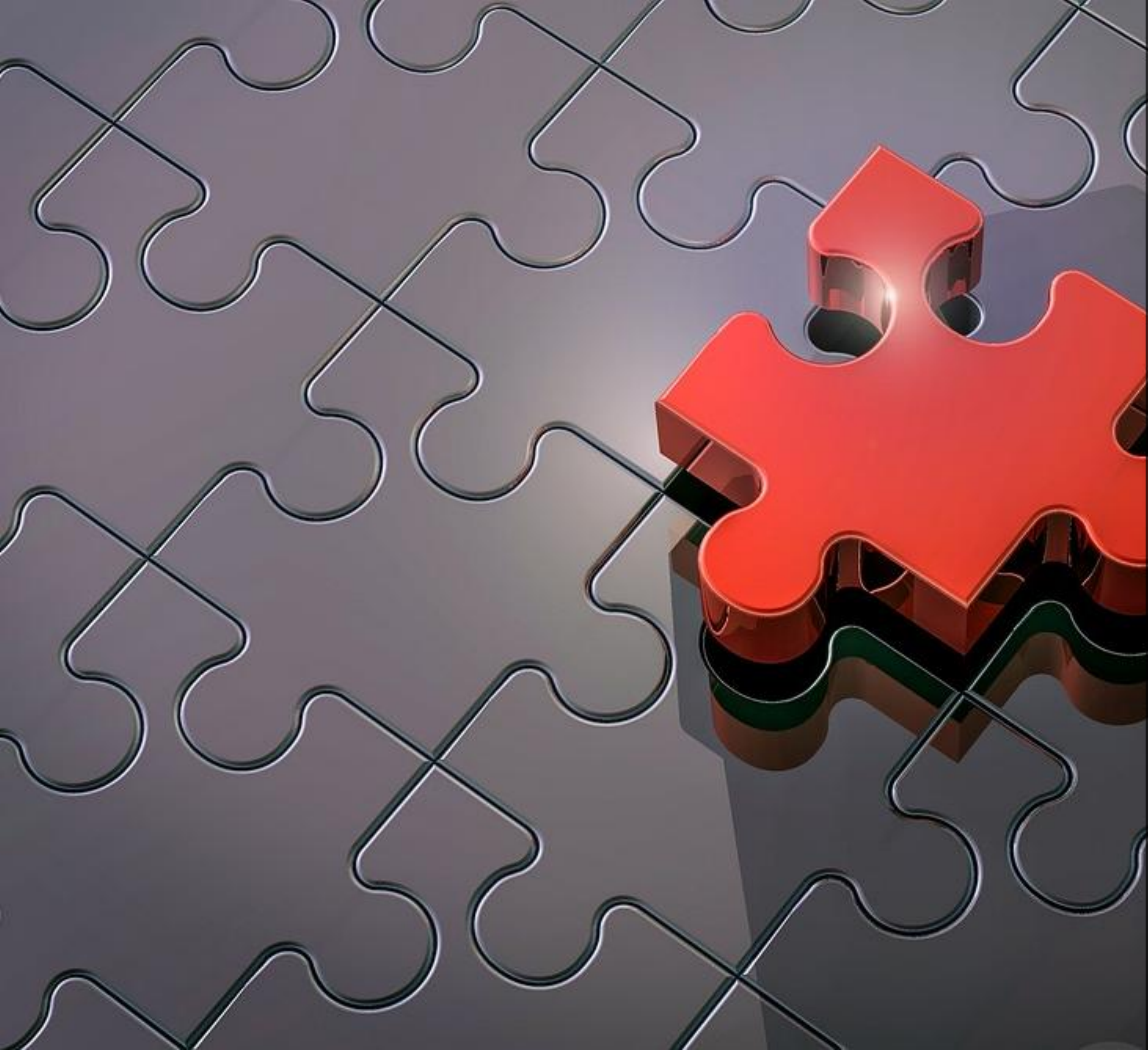
IT24100239 - Disanayaka D.M.C.N
IT24102276 - Kaldera H.P.I.D
IT24101177 - Bodini G.V.E.J
IT24102482 - Jayathilaka.D.L.T.S
IT24101092 - Perera .K.P.V.S
IT24101893 - Ranasinghe.N.A

Introduction

- Topic: **Home Autonomous System** for laundry management.
- Introducing **ClothMate**, an automated clothesline.
- Problem: Weather-related laundry issues.
- Solution: Smart clothesline adjusts automatically to weather.
- Features: Arduino-powered, smartphone control.
- Overview: Design, challenges, and benefits.

Session Map

- Introduction
- Problem Statement
- Objectives
- System Overview
- Components & Specifications
- System Design & Architecture
- Implementation Plan
- Evaluation Method
- Expected Outcomes
- Budget & Timeline
- Q&A

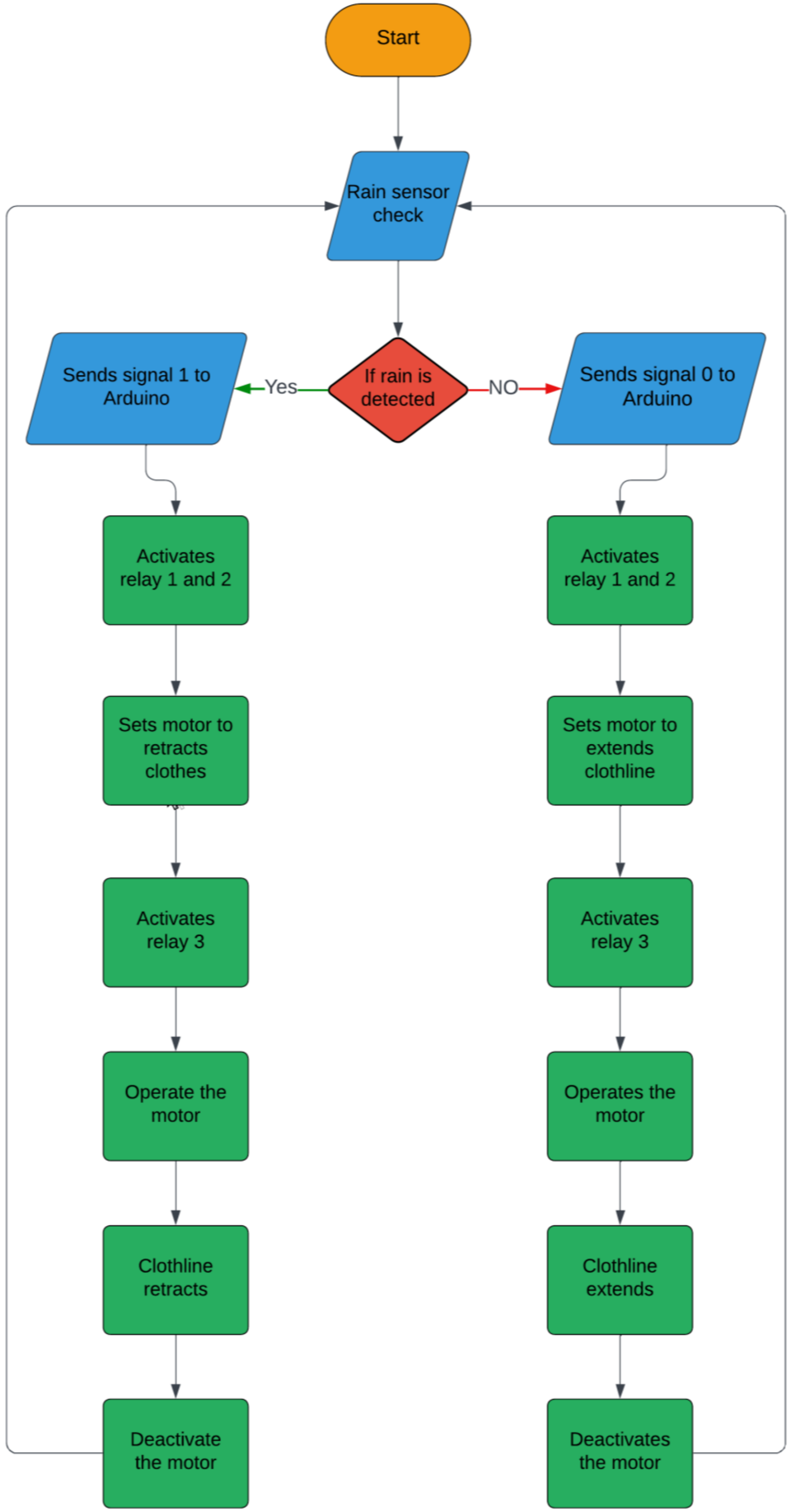


Problem Statement

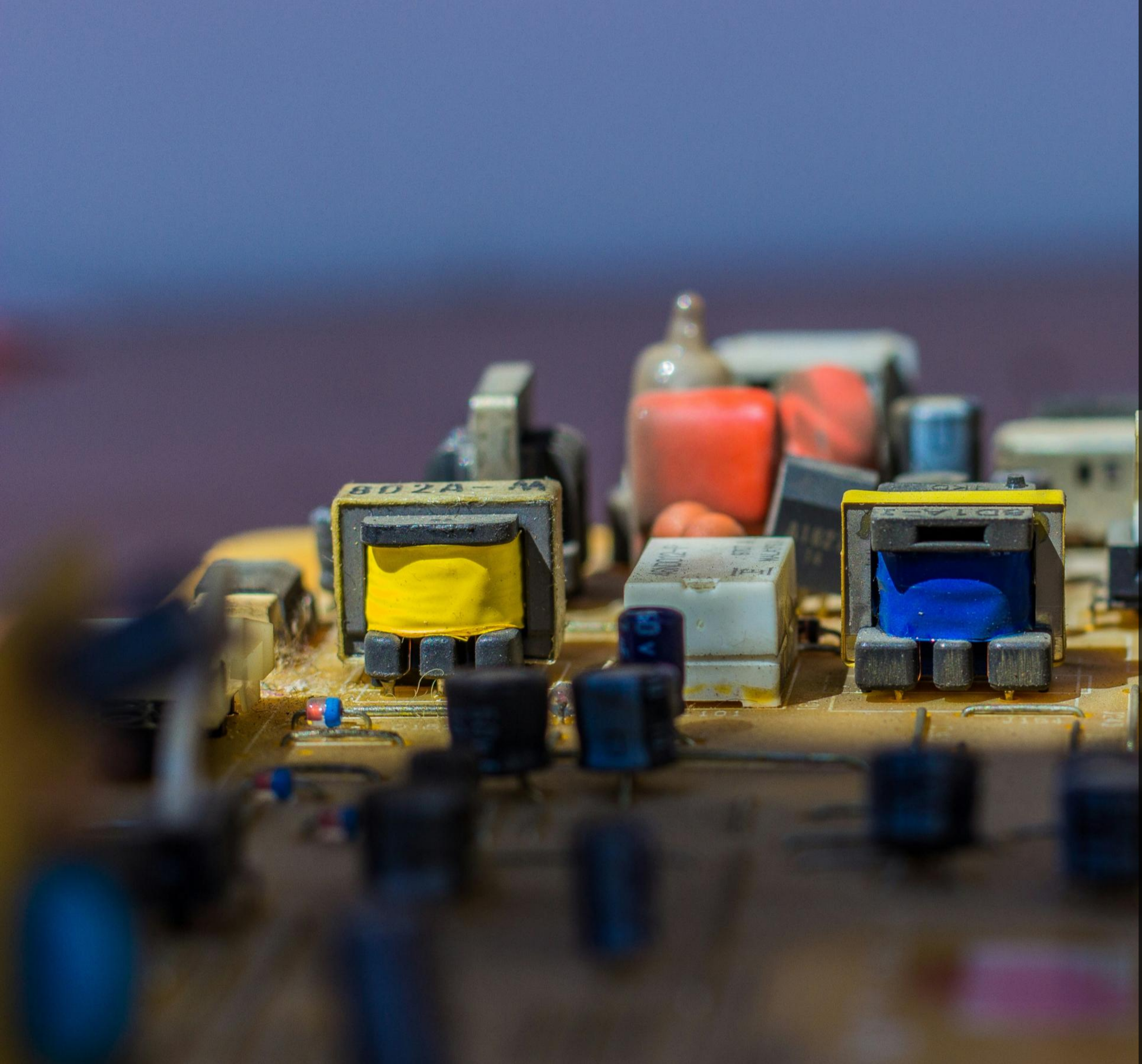
- Weather Unpredictability
- Manual Effort & Inconvenience
- Efficiency & Time Management
- Energy Consumption

Objective

- Project Planning
- Component Selection
- System Design & Prototyping



Start
Rain sensor check
If rain is detected
Yes
NO
Sends signal 1 to Arduino
Sends signal 0 to Arduino
Activates relay 1 and 2
Activates relay 1 and 2
Sets motor to retracts clothes
Sets motor to extends clothline
Activates relay 3
Activates relay 3
Operate the motor
Operates the motor
Clothline retracts
Clothline extends
Deactivate the motor
Deactivates the motor



System Overview

- Power on ⟶ Initialize system ⟶ Read Sensor Data

- Compare Weather Conditions

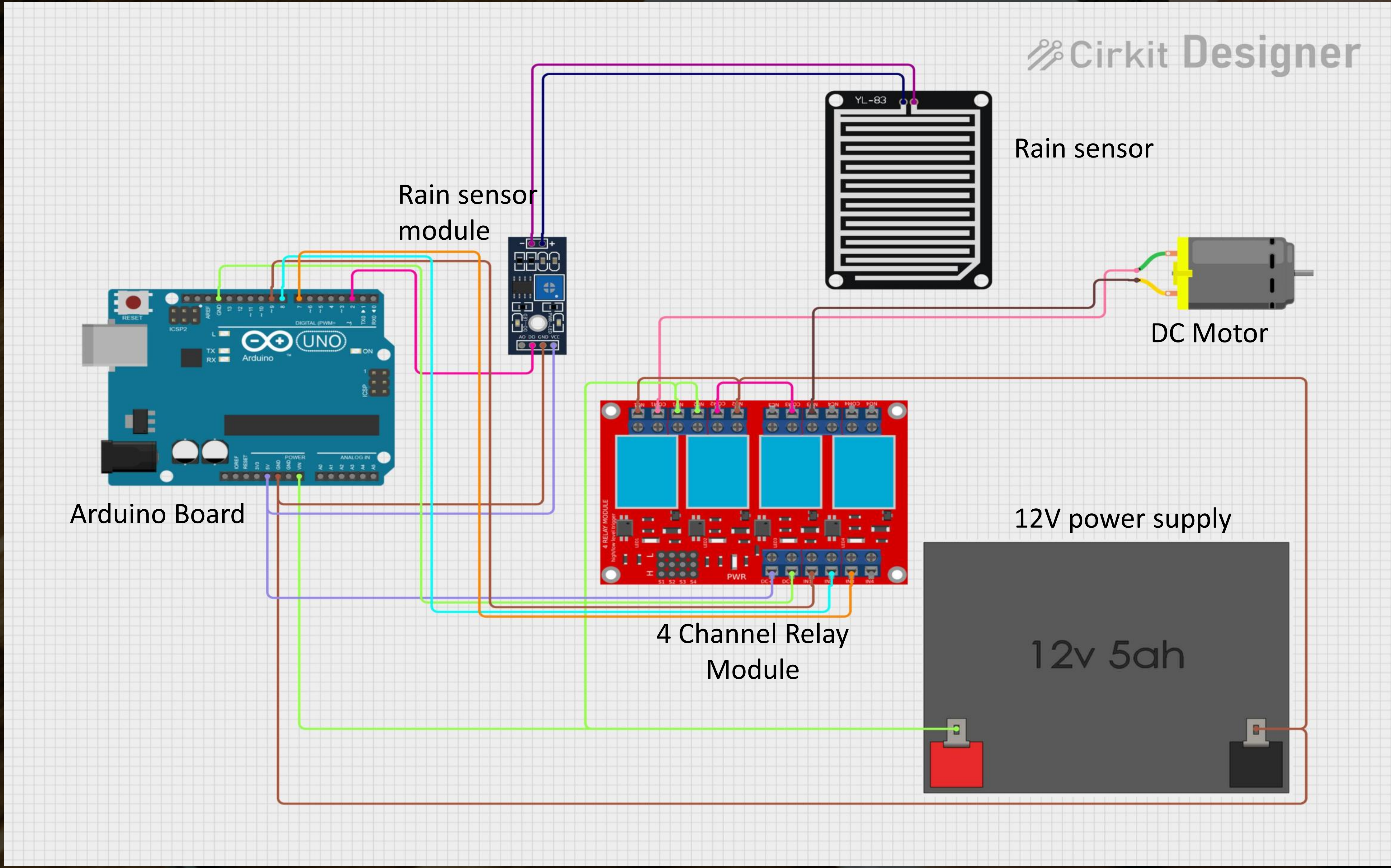
- **If Rain Detected** -> Retract Clothesline

- **If No Rain** -> Extend Clothesline/Check Mannual Commands

- Continue Monitoring

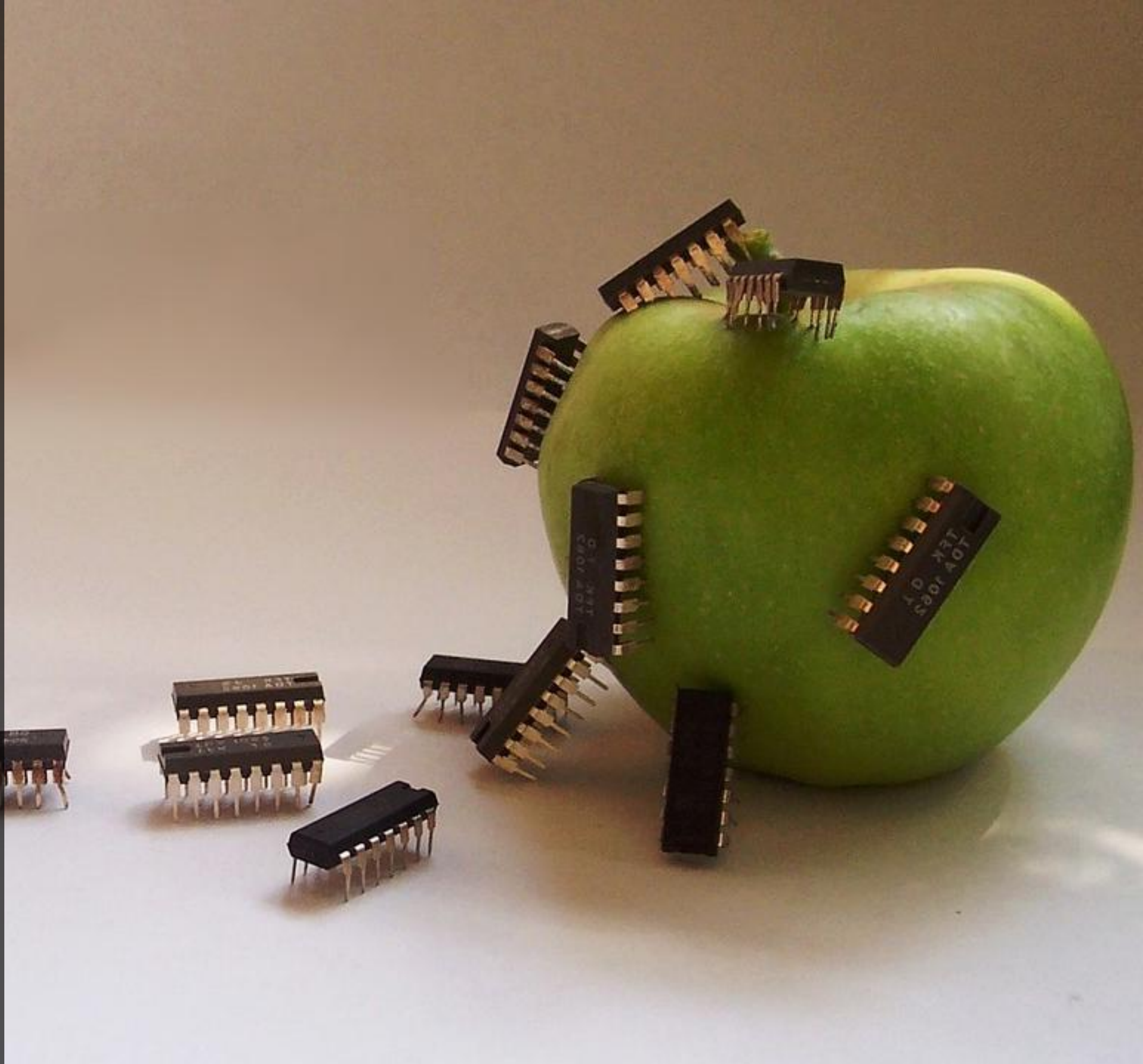
 - Deactivate Or Reactivate Based On Real Time Sensor Data

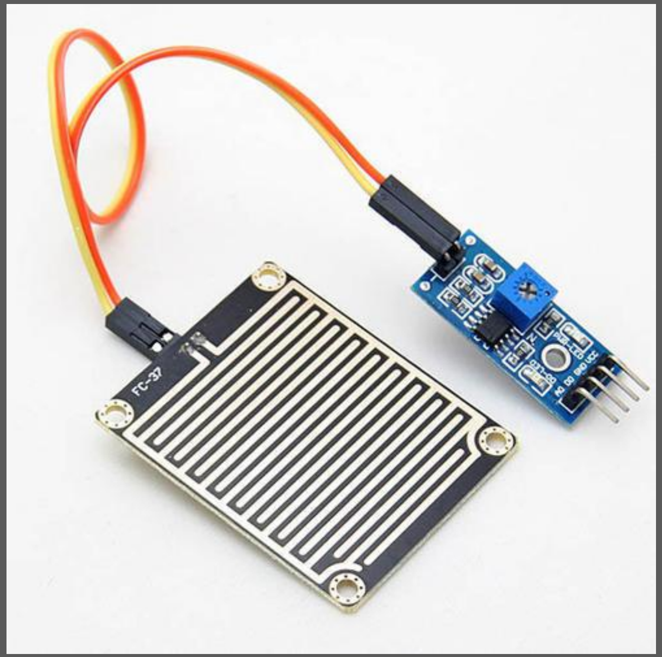
System Design & Architecture



Components

- Arduino Board
- Bread Board
- Jumper Wires
- Rain Sensor Module
- 4 Channel Relay Module
- DC Motor
- 12V Power Supply





Rain Sensor module

Arduino Board

Channel Relay module

DC motor

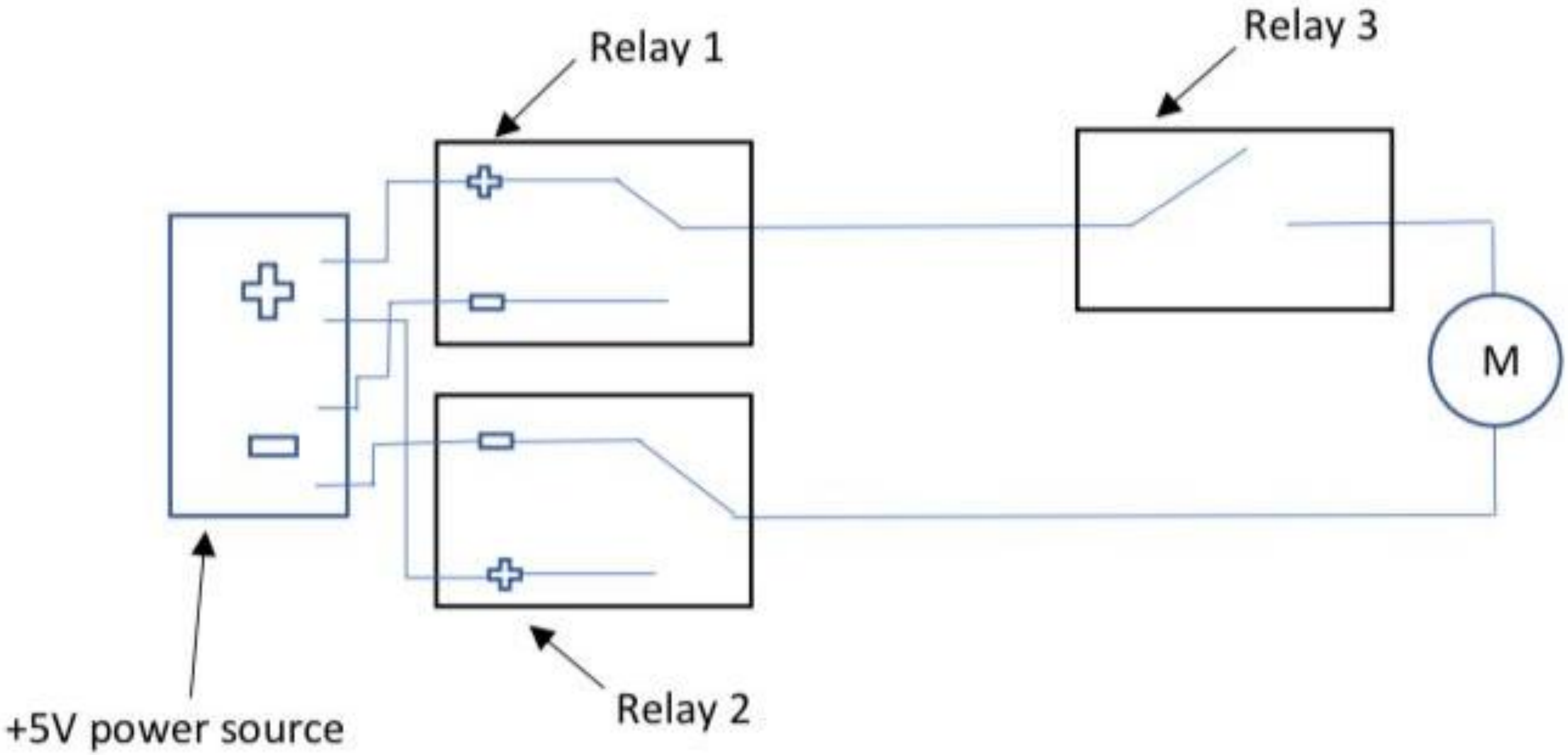
Implementation

1. **Rain Sensor:**

•Measures rainfall intensity via resistivity changes.
•Low resistance = High rainfall; High resistance = Low rainfall.

2. **Arduino Board:**
•Controls relays based on sensor input.
•Sends signal ‘1’ for low resistance, signal ‘0’ for high resistance.

3. **Relay Module:**
•Acts as a switch for motor operation.
•Handles high current & voltage, suitable for heavy loads.


Relay 1
Relay 3
M
Relay 2
+5V power source

Evaluation Method

- Test the accuracy of the rain sensor
- Check the relays & motor respond to the sensor
- Review the Arduino board

Expected Outcome

- Dry clothes
- Reliable Operation
- Smooth performance

Budget

- Arduino Board – Rs.1100
- Bread Board – Rs.140
- Jumper Wires – Rs.160
- Rain Sensor Module – Rs.160
- 4 Channel Relay Module – Rs.520
- DC Motor – Rs.500
- 12V Power Supply – Rs.1100

THANK YOU